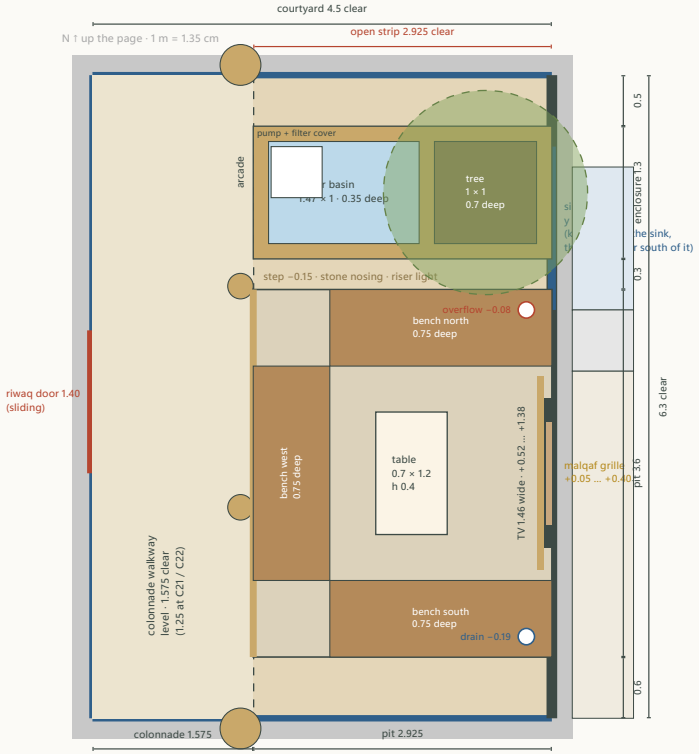
Courtyard layout · the room open to the sky · Taif / Palm House V7.1

Owner decisions 1 Nov 2026 (design review Y-02 / Y-03 / Y-04 / E-03) · clear dimensions between finished faces · the data in `3d/character.yaml` `courtyard_layout` and `studies/air_system.typ` (the grille) · nothing here is for construction



1 · Circulation and the real clear widths

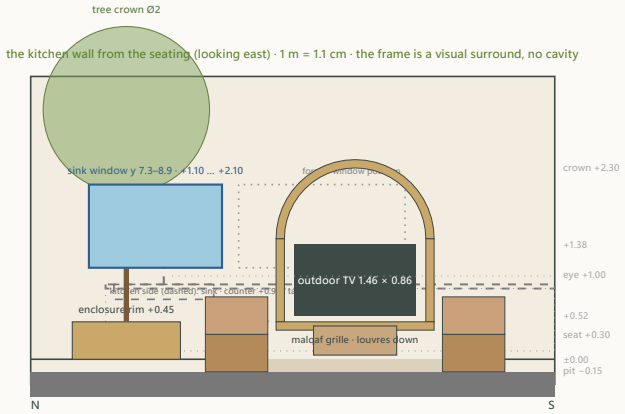
- **Courtyard:** 4.5 × 6.3 m between the finished faces (28.35 m²): the riwaq glass face x 4.35, the kitchen wall face x 8.85, the women's majlis glass y 6.6, the living glass y 12.9.
- **Colonnade walkway** under the projection, level, kept clear: 1.575 m from the riwaq glass to the soffit edge (x 5.925); 1.325 m at the two arcade columns (Ø0.25), 1.25 m at C21 / C22 (sleeves Ø0.4). The riwaq door (sliding, 1.40) opens onto it; it links the living's glass, the women's majlis glass and the riwaq.
- **Open strip east of the colonnade: 2.925 m** clear from the soffit edge to the kitchen wall (not 3.15 on axes; 2.85 m at the C21 / C22 sleeves).

2 · Seating one step down

- **The pit:** from the colonnade edge to the kitchen wall, 2.925 × 3.6 m (10.53 m²), floor −0.15, **one step** of 0.15 (the owner's choice); no walkway round it: the north and south strips are the cleaning strips at the glass (0.6 m at the living, 0.5 m at the women's majlis behind the enclosure) and the 0.3 m between the enclosure and the step.
- **Benches:** built-in, 0.75 deep (seat 0.45 above the pit floor = +0.30, back to +0.75), stone bases with cushions: west bench 2.1 m on the step edge facing the TV, with 0.75 m gaps at both ends to step down; north and south benches 2.17 m from the kitchen wall. Seat length 6.45 m: **4 comfortable with room to spare (7 places at 0.9 m each, 10 at 0.6 m)**; the east side stays free for the TV, the grille and cleaning (1.02 m).
- **Table:** 0.7 × 1.2 m, 0.4 high (meals), leg room 0.45 m west, 0.45 / 0.45 m north / south.
- **Step safety:** a contrasting honey-stone nosing 8 cm wide along the whole step edge, a continuous linear light (IP67 LED, warm) in the riser, no loose rugs on the edge; the bench backs (+0.75) edge the pit on three sides.

Study checks passed (10)

3 · The TV wall (the kitchen wall, seen from the seating) · 4 · The north enclosure: water and tree



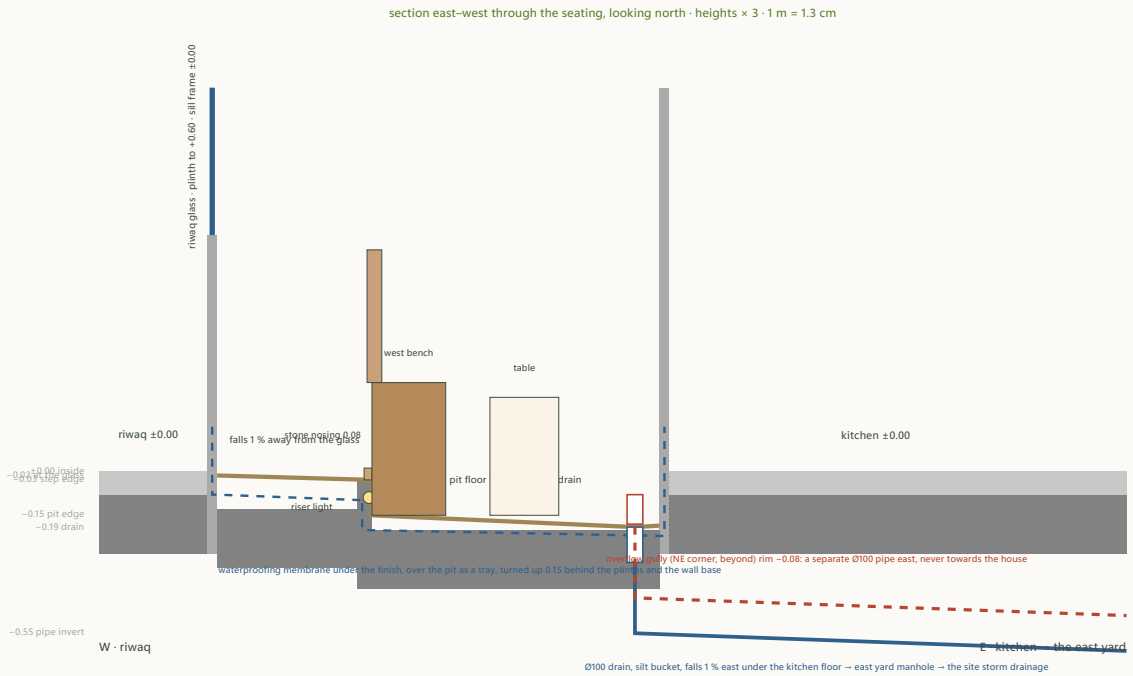
3 · The TV, its frame and the grille

- **The TV:** an **outdoor-rated** set (sun, rain and dust: an IP55 / IP56 “partial-sun” outdoor model, 65” class, ≈ 1.46 × 0.86 m; the final size, bracket, ventilation and installation clearances **per the chosen model**), centred on the seating at y 10.5, bottom +0.52 (0.67 above the pit floor), centre +0.95 against a seated eye at +1.00: the screen sits a little below eye level at 2.17 m from the west bench. Power and signal in a sleeved conduit from the kitchen; the fountain switch beside it.
- **The frame:** honey stone, a visual surround only: two jambs and a semicircular head, band 0.1 wide, projecting 0.06 from the wall, inner width 1.7 (0.12 each side of the TV), springing +1.45, crown +2.30 no tight cavity: the TV hangs on its bracket 0.06 m proud of the wall with the clearances the model asks (air behind and above it).
- **The grille** (`studies/air_system.typ` v3): below the TV, y 10–11, +0.05 … +0.40 (0.2 above the pit floor), low and decorative, removable for cleaning, with a **manually adjustable louvre** that sends the breeze along the floor to the benches and not into the faces; 1.6 m from the fountain enclosure’s splash; no bench or cushion in front of it (the east side of the pit stays free). The **damper switch** for the tower’s breeze path sits beside the TV with the fountain switch (the damper closes by itself when off or on power loss), and next to it the **assist fan’s own switch** (owner, 1 Nov 2026): a quiet EC fan box in the plenum directly behind the grille pulls the air down from the tower; it runs only with the damper open, and it slides out through the grille opening for service (no confined space). Behind the grille a fixed plenum under the sink counter (y 10–11) meets the breeze channel in the slab (10.5 m from the tower, cleaned from the laundry and kitchen floors).
- **The sink window, moved north** (owner, 1 Nov 2026): y 7.3–8.9 (1.6 m wide, sill +1.10, head +2.10) over the sink (y 7.5–8.9), the dishwasher south of it (y 8.9–9.5), the counter on to y 12.9 with the tap at the sink’s centre; the window looks at the fountain and the tree. Checks on the wall: 0.65 m of solid wall between the window’s south jamb and the TV’s frame; 0.2 m pier to the kitchen’s north wall; 0.55 m to column C26’s south face; the TV clears the window and the frame clears it. The kitchen side and the courtyard side of the wall are both in the elevation above (the kitchen’s counter line and sink drawn in grey on the same y axis).

4 · The north enclosure: one stone box, two basins

- **The box:** honey stone, x 5.93–8.85, y 7.1–8.4 (2.93 × 1.3 outside, walls 0.15, rim at +0.45 = a seat-height ledge), 0.5 m off the women’s majlis glass (cleaning), 0.3 m from the step.
- **Water basin** (west part): 1.47 × 1 inside, 0.35 deep, water at +0.25 a **laminar lip** in honey stone on the divider wall at +0.60, the water sheeting over stone into the basin (no spray, no mist); its own drain (bottom) and an overflow standpipe at the water level; the pump and the filter in a dry chamber under the access cover (0.5 × 0.5) at the west end; a **switch by the TV** turns the fountain off for watching; a GFCI / RCD circuit.
- **Planting basin** (east part, by the kitchen wall): 1 × 1 inside, 0.7 m of soil over a drainage layer and its own drain (adjust the depth to the chosen specimen); the basin tanked and separated from the water basin by the 0.15 divider; drip irrigation on a timer.
- **The tree:** medium, light canopy, bought at a good size, **fixed**: **Laurus nobilis** trained as a standard, or a clipped **Ficus nitida**; the final choice by a Taif nursery (climate, the winter cold, the irrigation water). Assumed: clear stem 2 m, crown Ø2 centred 3 m up, kept clipped off the kitchen wall and the glass.
- **Light check (simple):** the crown stands 1.15 m from the women’s majlis glass and covers 0.9 m of its 2.6 m width above +2.00: ≈ 11 % of that glass area in shade, the rest and the glass below +2.00 clear; the living’s glass is 5.15 m away: no shade. **To verify on the V7 3D model** (the sun paths) once the tree’s size is chosen.

6 · Drainage: the courtyard floor, the seating floor, the thresholds, the overflow and the discharge · section east–west through the seating (y ≈ 10.5)



The drainage, point by point

- **Thresholds:** the inside floors at ±0.00; the courtyard finish at the three glass lines and the riwaq door at **−0.02** (a 15 mm step, within the 15 mm of R-08); the glass sills on a plinth with the membrane turned up 0.15 behind it; a slot drain is not needed: the floor falls away from every door.
- **Falls:** the colonnade and the north / south strips fall 1 % away from the glass to the step edge (−0.03) and to the pit; the pit floor falls 1 % from its west edge (−0.15) to the **floor drain** at the south-east corner (−0.19, x 8.6, y 12.1): a Ø100 gully with a removable grating and a silt bucket, cleanable from the pit.
- **Overflow, independent:** a second gully at the north-east corner (x 8.6, y 8.9) with its rim at **−0.08**, 11 cm above the drain and below the step edge (−0.03): if the drain blocks, the pit fills to the overflow and the water leaves by a separate pipe east; it can never reach the glass lines, which stand 6.5 cm higher.
- **Pipes:** both run east under the kitchen floor (invert −0.55 at the drain, 1 % fall) across the kitchen (north of the dining door, south of the malqaf chamber) to the east wall and the **east yard manhole**, then to the **site storm drainage** (the plumbing strategy: everything east drains to the east yard; the plumbing study, open item 9, sets the manhole and the outfall). The kitchen's sealed malqaf cover is at y 2.9–3.7: the pipes pass north and south of the chamber, not through it.
- **The fountain:** the water basin's bottom drain (−0.35) and its overflow standpipe (at the water level +0.25) join the overflow pipe; the planter's drain (bottom −0.70) joins it too, through a gravel layer and a geotextile; the pump chamber has its own small drain. The water basin is filled from a tap in the pump chamber; the fountain runs on a timer and the switch by the TV.
- **Waterproofing:** one membrane over the whole courtyard slab (lowered 0.05 for the outdoor build-up), over the pit as a tray with the step as an upstand, up 0.15 behind the glass plinths and the kitchen wall base; the enclosure's two basins tanked separately; the finish honey-light stone (slip-resistant outdoors), the step nosing honey stone.
- **Engineer / MEP to confirm:** the slab recess for the pit (0.15) and the planter (0.75), the gully and pipe sizes, the east yard manhole and the outfall.

Dimensions (clear, between finished faces)

Item	m	Note
Courtyard clear	4.5 × 6.3	28.35 m²; axes 4.70 × 6.50
Colonnade walkway (level)	1.575 / 1.325 / 1.25	to the soffit edge / at the arcade columns / at C21–C22
Open strip east of the colonnade	**2.925**	the soffit edge x 5.925 → the kitchen wall face x 8.85 2.85 at the sleeves
Seating pit	2.925 × 3.6	10.53 m², floor −0.15, one step 0.15
Benches	0.75 deep, 6.45 long	seat +0.30, back +0.75 7 comfortable, 10 max
Table	0.7 × 1.2 × 0.4	leg room 0.45 W, 0.45 N, 0.45 S; 1.02 free to the wall
Cleaning strips	0.6 S · 0.5 N	at the living glass / behind the enclosure at the women's majlis glass; 0.3 enclosure → step
Step	0.15	nosing 0.08 honey stone, riser light, no rugs
TV	1.46 × 0.86	y 9.77–11.23, +0.52 … +1.38 per the chosen model
TV frame	1.7 inside, band 0.1	springing +1.45, crown +2.30, proud 0.06
Grille	1 × 0.35	y 10–11, +0.05 … +0.40 0.12 below the TV

Enclosure	2.93 × 1.3	walls 0.15, rim +0.45
Water basin	1.47 × 1 × 0.35	water +0.25, lip +0.60
Planting basin	1 × 1 × 0.7	tree crown Ø2 at 3 m
Levels	−0.02 / −0.03 / −0.15 / −0.19	glass line / step edge / pit edge / drain; overflow rim −0.08 pipe invert −0.55

Conflicts and decisions

- **The kitchen sink window** is moved north to y 7.3–8.9 (owner, 1 Nov 2026): the TV and its frame stay where they are, 0.65 m of solid wall between them and the window (page 2).
- **The 3D character data** now carry the new layout (`courtyard_layout`); the builder (`character_3d.py`) still draws the old sunken seating, fountain and pergola until it is updated at the next 3D build.
- **The pergola** is removed from the design (owner): the courtyard enclosure study's note and the character data; the roof keeps its guards.
- **Children's play** moves to the garden (R-23 / R-30 updated); the courtyard is for sitting, meals and the TV.